11. Simulation and Performance Evaluation of OFDM and FBMC Wireless Waveforms

MATLAB Program Script

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clc; clear; close all;
%% 1. Input Parameters Setup
N = 64; % Total subcarriers
M = 12; % Active subcarriers (centered)
N_cp = 16; % Cyclic prefix for OFDM
K_fbmc = 4; % Overlapping factor for FBMC
N_sym = 100; % Number of symbols to simulate
rng(42); % Set seed for reproducible results
% Active subcarrier index allocation
active_idx = (N/2 - M/2 + 1) : (N/2 + M/2);
%% 2. Data Generation and QAM Mapping
num_bits_per_sym = M * 2; % 4-QAM = 2 bits/symbol
data_bits = randi([0 1], num_bits_per_sym, N_sym);
% Modulate bits to QAM symbols
qam_mod = zeros(M, N_sym);
for s = 1:N_sym
bits = data_bits(:, s);
for m = 1:M
b_idx = (m-1)*2 + 1;
% Simple QAM Gray Mapping
real_part = 1 - 2*bits(b_idx);
imag_part = 1 - 2*bits(b_idx+1);
qam_mod(m, s) = (real_part + 1i*imag_part)/sqrt(2);
end
end
%% 3. OFDM Waveform Generation
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ofdm_tx_sig = [];
for s = 1:N_sym
ifft_in = zeros(N, 1);
ifft_in(active_idx) = qam_mod(:, s);
time_sym = ifft(ifft_in) * sqrt(N);
% Append Cyclic Prefix
cp_part = time_sym(end - N_cp + 1:end);
ofdm_tx_sig = [ofdm_tx_sig; cp_part; time_sym];
end
%% 4. FBMC Waveform Generation (OQAM Processing)
% PHYDYAS Prototype Filter Coefficients for K=4
H1 = 0.97195983; H2 = 0.70710678; H3 = 0.23514752;
g = zeros(1, 4*N);
for t = 0:(4*N-1)
g(t+1) = 1 - 2*H1*cos(2*pi*t/(4*N)) + 2*H2*cos(4*pi*t/(4*N)) -
2*H3*cos(6*pi*t/(4*N));
end
g = g / sqrt(sum(g.^2)); % Normalize filter energy
fbmc_tx_sig = zeros(N*N_sym + 4*N, 1);
% OQAM Implementation via synthesis equation scaling
for s = 1:N_sym
for m = 1:M
k = active_idx(m);
sym_data = qam_mod(m, s);
% Split into In-phase (Real) and Quadrature (Imag) components
r_data = real(sym_data);
i_data = imag(sym_data);
% Phase tracking offsets for orthogonality constraint
phase_r = exp(1i * pi * (k-1) * (0 - (4-1)/2));
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phase_i = exp(1i * pi * (k-1) * (N/2 - (4-1)/2)) * 1i;
% Time domain modulation & windowing arrays
t_arr = 0:(4*N-1);
carrier_r = exp(1i * 2 * pi * (k-1) * t_arr / N) * phase_r;
carrier_i = exp(1i * 2 * pi * (k-1) * t_arr / N) * phase_i;
shift_r = (s-1)*N + 1;
shift_i = (s-1)*N + N/2 + 1;
fbmc_tx_sig(shift_r : shift_r + 4*N - 1) = fbmc_tx_sig(shift_r : shift_r + 4*N - 1) +
(r_data * g .* carrier_r).';
fbmc_tx_sig(shift_i : shift_i + 4*N - 1) = fbmc_tx_sig(shift_i : shift_i + 4*N - 1) +
(i_data * g .* carrier_i).';
end
end
%% 5. Metric Calculations (PAPR)
window_len_ofdm = N + N_cp;
ofdm_papr_vals = zeros(N_sym, 1);
for s = 1:N_sym
seg = ofdm_tx_sig((s-1)*window_len_ofdm + 1 : s*window_len_ofdm);
ofdm_papr_vals(s) = max(abs(seg).^2) / mean(abs(seg).^2);
end
max_ofdm_papr = 10 * log10(max(ofdm_papr_vals));
fbmc_papr_vals = zeros(N_sym, 1);
for s = 1:N_sym
seg = fbmc_tx_sig((s-1)*N + 1 : (s-1)*N + N);
if mean(abs(seg).^2) > 0
fbmc_papr_vals(s) = max(abs(seg).^2) / mean(abs(seg).^2);
end
end
max_fbmc_papr = 10 * log10(max(fbmc_papr_vals));
```

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%% 6. Display Performance Metrics
fprintf('\nWAVEFORM COMPARISON SIMULATION RESULTS\n');
fprintf('=======================================\n');
fprintf('OFDM Max Inst. PAPR : %.3f dB\n', max_ofdm_papr);
fprintf('FBMC Max Inst. PAPR : %.3f dB\n', max_fbmc_papr);
%% 7. Plotting Power Spectral Density (PSD)
[psd_ofdm, f_ofdm] = periodogram(ofdm_tx_sig, [], 2048, 1, 'centered');
[psd_fbmc, f_fbmc] = periodogram(fbmc_tx_sig, [], 2048, 1, 'centered');
figure('Name', 'Waveform Spectral Evaluation');
plot(f_ofdm, 10*log10(psd_ofdm/max(psd_ofdm)), 'r-', 'LineWidth', 1.2); hold on;
plot(f_fbmc, 10*log10(psd_fbmc/max(psd_fbmc)), 'b-', 'LineWidth', 1.2); grid on;
xlabel('Normalized Frequency (\times \pi rad/sample)');
ylabel('Relative Power Density (dB)');
title('PSD Comparison: OFDM vs. FBMC');
legend('OFDM (Rectangular Window)', 'FBMC (PHYDYAS Prototype Filter)',
'Location',
'southoutside');
ylim([-80 5]);
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Editor

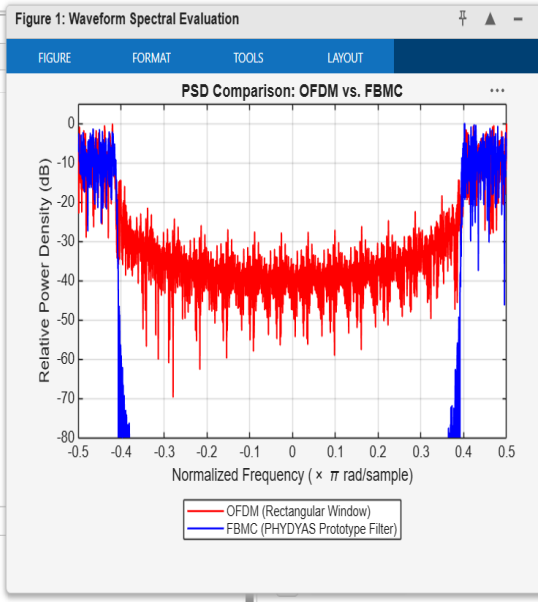
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71          % Split into In-phase (Real) and Quadrature (Imag) components
72          r_data = real(sym_data);
73          i_data = imag(sym_data);
74
75          % Phase tracking offsets for orthogonality constraint
76          phase_r = exp(1i * pi * (k - 1) * (0 - (K_fbmc - 1) / 2));
77          phase_i = exp(1i * pi * (k - 1) * (N / 2 - (K_fbmc - 1) / 2)) * 1i;
78
79          % Time domain modulation & windowing arrays
80          t_arr = 0:(4 * N - 1);
81          carrier_r = exp(1i * 2 * pi * (k - 1) * t_arr / N) * phase_r;
82          carrier_i = exp(1i * 2 * pi * (k - 1) * t_arr / N) * phase_i;
83
84          shift_r = (s - 1) * N + 1;
85          shift_i = (s - 1) * N + N / 2 + 1;
86
87          fbmc_tx_sig(shift_r:shift_r + 4 * N - 1) = fbmc_tx_sig(shift_r:shift_r + 4 * N -
88              (r_data * g .* carrier_r).';
```

Command Window

```
WAVEFORM COMPARISON SIMULATION RESULTS
=======================================
OFDM Max Inst. PAPR : 9.690 dB
FBMC Max Inst. PAPR : 9.971 dB
>>
```



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Validate generated output before use.